

Multi-scale pyramid-former network with multiple consistency constraints for semi-supervised video action detection

Qiming Zhang^{a,*}, Zhengping Hu^{a,b,*}, Yulu Wang^a, Hehao Zhang^a, Jirui Di^a

^a School of information and engineering & Yanshan University, Hebei Qinhuangdao 066004 China

^b Yanshan University & Hebei Key Laboratory of information transmission and signal processing, Hebei Qinhuangdao 066004 China

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ABSTRACT

Current semi-supervised video action detection methods predominantly emphasize consistency regularization across data augmentations, while overlooking cross-scale consistency modeling in unlabeled video data. To address this limitation, this paper proposes the Multi-Scale Pyramid-Former Network with multiple consistency constraints, termed MSPF Net. Specifically, MSPF Net employs a novel Pyramid Fusion Strategy to integrate action representations at the current scale with those from other scales through weighted fusion. This fusion strategy is embedded in each layer of MSPF Net, with each layer representing a different scale. Then, MSPF Net aggregates representations from different layers to maximize the extraction of scale information from action descriptors in unlabeled videos. Moreover, this paper employs a multiple consistency strategy to impose constraints on multi-scale information in MSPF Net, thereby further enhancing model performance. Experiments were conducted on the JHMDB-21 and UCF101-24 datasets, and the results demonstrated that MSPF Net achieved a 3.1 % and a 0.9 % improvement over the state-of-the-art methods in terms of v-mAP@0.5 on the two datasets, respectively. Furthermore, the visualization results provide additional evidence that MSPF Net can accurately focus on action instances even in the absence of labels.

1. Introduction

Driven by the proliferation of internet technologies, the growing demand for video recording and accelerated transmission speeds have led to the accumulation of massive video datasets encompassing diverse scenarios and human actions. In response to these technological trends, video understanding algorithms have emerged as a critical research focus, with increasing attention being paid to addressing real-world challenges in multimedia analysis. Video action classification, a pivotal research domain in video understanding, has achieved significant advancements through the synergistic evolution of convolutional architectures and deep learning frameworks. However, the effectiveness of video action classification depends on the utilization of large-scale datasets (Kay et al., 2017; Soomro et al., 2012). Compared to video action classification, video action detection (VAD) is more challenging because it not only requires predicting the categories of actions in the video, but also their spatiotemporal localization. Correspondingly, preparing large-scale datasets for video action detection is even more difficult, as it requires not only video-level category annotations but also frame-level localization annotations, a process that undoubtedly demands significant time and manpower.

Due to the limitations of training conventional deep learning models and the urgent need to reduce data labeling costs, there has

* Corresponding authors.

E-mail addresses: zhangqiming@stumail.ysu.edu.cn (Q. Zhang), hzp@ysu.edu.cn (Z. Hu).

been a growing focus on semi-supervised learning (SSL). Unlike fully supervised learning approaches, semi-supervised learning methodologies can effectively train models using minimal labeled datasets while leveraging abundant unlabeled data. This can not only effectively reduce the difficulty and cost of annotation, but also enable model training by fully utilizing unlabeled data, thereby effectively mitigating overfitting in data-limited scenarios. When dealing with a new action detection task, it is relatively easy to train the model using annotated data because neural networks can easily locate and extract corresponding action features based on the ground truth of each frame of the video. However, effectively leveraging unlabeled data for training in the absence of labels presents a greater challenge. This is primarily because neural networks struggle to distinguish between action foreground and irrelevant background in unlabeled data, which is a key factor contributing to the low mean average precision (mAP) in semi-supervised video action detection (SSVAD).

In order to improve the utilization efficiency of unlabeled data under semi-supervised learning conditions, it is crucial to establish effective consistency constraints on the unlabeled data. Therefore, previous methods have proposed a series of constraint methods based on consistency regularization. For example, CutMix-CD (Shu et al., 2025) develops separate Student and Teacher paths and imposes consistency constraints on their prediction results. VSSL (Wang et al., 2025) extracts spatiotemporal features of actions from both the original and augmented videos, then enforces consistency between their predictions. Although these methods have achieved good results, they generally impose consistency constraints from a low-level shallow perspective. To fully explore consistency across different scales and fine-grained features in unlabeled video data, and to establish consistency constraints between modalities of high-resolution action localization features, this paper proposes the Multi-Scale Pyramid-Former Network (MSPF Net), which leverages semi-supervised consistency regularization to enhance performance. Specifically, to improve the consistency of different scale features between unlabeled original and augmented datasets in semi-supervised learning, this paper proposes a Coarse-to-Fine Multi-Scale Contrastive Strategy to apply consistency constraints on the features of different layers in MSPF Net. In order to better distinguish between primary action areas and irrelevant background in videos, this paper imposes consistency constraints on the high-resolution action localization features output by the model and proposes a Pixel Domain Stepwise Localization Refinement Strategy. In the Pixel Domain Stepwise Localization Refinement Strategy, action foreground regions are first detected using action localization features. A threshold-activated neighboring frame action migration constraint is then applied to suppress dominant background regions and activate action foregrounds, while further refinement of action boundaries is achieved by calculating gradients on the migration map, ultimately enhancing action localization accuracy.

In summary, the primary contributions of this study are delineated as follows:

1. This paper proposes the Multi-Scale Pyramid-Former Network (MSPF Net) based on semi-supervised learning. First, we utilize the Pose Encoder and Pose Activator modules to perform feature enhancement and capture contextual information in the video. Then, we perform multi-scale fusion of high-resolution global features, local detail features, and low-resolution semantic features from residual blocks to enhance the model's ability to perceive unlabeled data. To our knowledge, this is the first multi-scale pyramid-former structure for semi-supervised video action detection.
2. This paper introduces a Coarse-to-Fine Multi-Scale Contrastive Strategy that is applied to multi-scale and multi-resolution features from the MSPF Net layers, enhancing the model's ability to capture information across varying scales.
3. To address blurred action boundaries caused by complex backgrounds, this paper proposes a Pixel Domain Stepwise Localization Refinement Strategy that refines detection boundaries using consistency regularization across multiple modalities in the pixel domain.
4. Extensive comparative and ablation experiments were conducted on benchmark datasets, demonstrating that the proposed approach outperforms existing state-of-the-art techniques with superior performance metrics.

2. Related works

2.1. Video action detection

VAD not only requires predicting the categories of actions in the video, but also their localization. Due to the intricate nature of video data and the scarcity of annotated information, most methods often use a dual stream structure to separately process spatial features and temporal features, which is relatively complex in structure. In response to the above issue, Hou et al. (2017) introduce an efficient end-to-end framework utilizing Tube Convolutional Neural Networks (T-CNN) for detection tasks, which significantly streamline the model's architectural complexity. Similarly, in response to the problem of large-scale and heavy computation in dual stream network models, Zhao and Snoek (2019) successfully reduce the computational and parameter requirements to half of the state-of-the-art dual stream methods by embedding RGB information and optical flow information into a two-in-one stream network. In order to improve the performance of anchor free action detectors, Ma et al. (2021) propose a Self-Attention Moving Center Detector (SAMOC) that enhances representation by aggregating global context information. Similar to SAMOC, the Cascaded Spatial-Temporary Attention network (CSTA Net) (Yang et al., 2023) uses an attention mechanism to introduce a Spatial-Aware Attention Module (SAM) and a Temporal-Aware Attention Module (TAM), effectively enhancing the detector's ability to track activities. To prevent the loss of crucial spatial semantic information during the detection process, Sem-Adapter (Zhao et al., 2025) designs the Self-Motion Module (SMM) and the Mutual Relation Module (MRM) respectively to focus on semantic information and model the interactions between semantic features.

2.2. Semi-supervised learning

SSL minimizes the expense associated with data annotation by leveraging a limited set of labeled samples alongside a substantial volume of unlabeled data to train the model. The existing semi-supervised learning methods can be mainly divided into two categories: consistency regularization methods and pseudo-labeling methods. The pseudo-labeling method first trains the model using labeled data, then generates pseudo-labels by predicting unlabeled data, and finally combines the original labeled data with these new pseudo-labels for joint model training. Lee (2013) first proposes the pseudo-labeling approach, where the one-hot labels generated by the model are used as pseudo-labels for unlabeled samples and utilized in model training. To address the issue that different categories of features pose varying levels of difficulty for model learning, ATD-SS (Wang et al., 2025) proposes an adaptive threshold-driven pseudo-labeling method. This method adaptively adjusts the threshold according to the predicted probability distribution, enabling more efficient use of the data. The consistency regularization approach introduces specific perturbations to the input data, aiming to ensure that the model's predictions on the perturbed data align with those on the unaltered data. In object detection tasks, the main background often causes differences in model consistency discrimination. To address the above issue, Co-SSD (Jeong et al., 2019) defines masks for background elimination, reducing the influence of non-detection areas and thereby enhancing the model's focus on the target. Berthelot et al. (2019) unify the previous mainstream methods of semi-supervised learning and propose a new algorithm called MixMatch based on regularization method, minimum entropy method, and Mixup method, which greatly improving model performance. To address the issue that unidirectional regularization can lead to disturbances in prediction noise, Cao et al. (2025) mitigate the impact of high-confidence incorrect predictions by re-weighting pixel-level predictions and applying supplementary regularization. To overcome the limitation that existing methods overlook feature channel information, Li et al. (2025) combine the graph convolutional network with a semi-supervised consistency framework and impose channel-level constraints between different decoders, markedly boosting model performance.

2.3. Semi-supervised video action detection

To reduce the high annotation cost in video action detection and improve the model's adaptability across different scenarios, international scholars have investigated this challenge using semi-supervised learning approaches. The current SSVAD methods mainly rely on the consistency regularization method, as pseudo-labeling is an iterative approach requiring continuous iterations to generate pseudo-labels, thereby incurring high computational costs, whereas consistency regularization avoids multiple iterations. Kumar and Rawat (2022) propose an effective end-to-end SSVAD model based on the consistency regularization method. Due to the tendency of the model to recognize dynamic backgrounds in videos as foreground actions, BWCC (Zhong et al., 2023) build upon the work of Kumar and Rawat (2022) and introduce a Calibration Constraint to ensure high consistency in the model's predictions within video action regions. In order to enhance the information content of labeled samples, Singh et al. (2024) incorporate active learning into a semi-supervised framework, thus further improving model performance by actively selecting samples with higher informational value. To enhance the model's ability to distinguish between actions and backgrounds in videos, Multi-FB (Zhang et al., 2025) compares the action foreground and background of unlabeled videos at different granularities, allowing the model to effectively differentiate irrelevant backgrounds in unannotated scenarios.

2.4. Multi-scale pyramid network structure

The multi-scale pyramid network structure can effectively capture feature information at different scales, thereby improving model performance. Therefore, many researchers have conducted studies on the multi-scale pyramid network structure. U-Net (Ronneberger et al., 2015) extracts pyramid features at different scales through consecutive encoders and connects these pyramid features through skip connections to achieve better segmentation details. Pyramid in Transformer (PiT) (Zhang et al., 2022) combines directional and multi-scale features to form pyramid features for each pedestrian image, then trains on these features to effectively improve pedestrian retrieval accuracy. To address the challenge of representing multi-scale information in remote sensing images using a single-scale design, Xiao et al. (2022) propose a joint framework based on a multi-scale spatial-temporal transformer. This framework embeds information from different scales into self-attention mechanisms, effectively enhancing the model's feature representation capability. To extract more discriminative fine-grained and coarse-grained spatial features, MAST (Wang et al., 2025) generates features at different scales through multi-scale pooling and then uses a spatial transformer to capture the relationships among these multi-scale features. Although the multi-scale pyramid network structure has advanced in many research fields, it has seen limited progress in SSVAD. The main reason is that the multi-scale pyramid structure requires a large amount of annotated data for model training. In semi-supervised learning, the number of labeled samples is limited, leading to poor robustness and overfitting when the model handles unlabeled data. To address this issue, this paper imposes consistency constraints on our proposed MSPF Net through the Coarse-to-Fine Multi-Scale Contrastive Strategy and the Pixel Domain Stepwise Localization Refinement Strategy, effectively enhancing the model's performance in the absence of labeled data. Furthermore, compared to fusing multi-scale information via concatenation, this paper proposes a Pyramid Fusion Strategy to aggregate information of different scales from different layers in the MSPF Net, allowing the model to focus more precisely on the key areas within the video.

3. Methodology

This part provides a detailed explanation of our novel Multi-Scale Pyramid-Former Network. In Section 3.1, we outline the research

objectives of this study. In Section 3.2, we present an overview of the network's overall architecture. In Section 3.3, we describe the implementation of the Pose Encoder and Pose Activator modules, which are designed to enhance features and capture contextual information from videos. In Section 3.4, we elaborate on the Multi-Scale Pyramid-Former Network and its associated Pyramid Fusion Strategy, which is designed to enable multi-scale feature fusion. In Section 3.5, we detail the multiple consistency constraints employed in our approach.

3.1. Research objective

The objective of this study is to train our proposed MSPF Net using only a limited number of annotated samples and apply it to video action detection tasks. For the input data, we divide it into a labeled video set $V_L = \{v_1^l, v_2^l, \dots, v_m^l\}$ and an unlabeled video set $V_{UL} = \{v_1^{ul}, v_2^{ul}, \dots, v_n^{ul}\}$. Each annotated video contains action category labels y_i and localization labels X_i . The objective of MSPF Net is to jointly leverage the labeled set V_L and the unlabeled set V_{UL} to effectively extract multi-scale action descriptors from videos for pyramid fusion, thereby enabling richer feature representation and further improving the model's detection accuracy under semi-supervised conditions. Details regarding the structure and implementation of MSPF Net will be presented in the following sections.

3.2. Overview of the model

In semi-supervised learning, our goal is to extract richer modal information from unlabeled data for action discrimination and consistency regularization constraints. For input videos $X = \{X_1, X_2, \dots, X_n\}$ which contain both unlabeled and labeled data, we adjust the resolution of all videos to 224×224 to facilitate multi-resolution processing of the model. The overall overview of the proposed Multi-Scale Pyramid-Former Network is shown in Fig. 1. Firstly, the I3D backbone network (Carreira and Zisserman, 2017) is pre-trained on the Kinetics dataset (Kay et al., 2017) to equip it with the ability to extract basic features. Then we input the video data $\{X_i\}$ into the I3D network and use the feature $F_{PH} \in \mathbb{R}^{B \times C \times T \times H \times W}$ output by the first block in the backbone as a primary high-resolution global feature, where B represents batch size, C represents the number of channels, T represents the number of video frames and is set to 8, H and W represent feature height and width and are equal to 112. We also take the feature $F_{PL} \in \mathbb{R}^{B \times C \times T \times H \times W}$ output from the fourth block of the backbone as the primary local detail feature, where H and W are equal to 56. We also take the feature $F_{PA} \in \mathbb{R}^{B \times C \times T \times H \times W}$ output from the last block of the backbone as the primary low-resolution abstract semantic feature, where H and W are equal to 28. Then input F_{PA} sequentially into the pre-processing, Pose Encoder and Pose Activator modules to perform feature enhancement and capture contextual information from the video. Detailed information about the Pose Encoder and Pose Activator can be found in Section 3.3. Then we input the primary high-resolution global feature F_{PH} , primary local detail feature F_{PL} , and primary low-resolution abstract semantic feature F_{PA} as multi-scale residual information into the MSPF Net. After applying the Pyramid Fusion Strategy, the MSPF Net outputs the multi-scale pyramid features $\{M_1, M_2, M_3\}$, where $M_1 \in \mathbb{R}^{B \times C \times T \times 28 \times 28}$, $M_2 \in \mathbb{R}^{B \times C \times T \times 56 \times 56}$, $M_3 \in \mathbb{R}^{B \times C \times T \times 112 \times 112}$ corresponding to each layer. The last layer of the MSPF Net outputs the action localization feature $M_{loc}^- \in \mathbb{R}^{B \times C \times T \times 112 \times 112}$ that integrates multi-scale residual information. After post-processing, the high-resolution action localization feature $M_{loc} \in \mathbb{R}^{B \times C \times T \times 224 \times 224}$ is obtained. It is worth noting that this study uses multi-scale pyramid features $\{M_1, M_2, M_3\}$ for the Coarse-to-Fine Multi-Scale Contrastive Strategy, and the high-resolution action localization feature M_{loc} for the Pixel Domain Stepwise Localization Refinement Strategy. These strategies are based on consistency regularization methods, which jointly constrain the MSPF Net to minimize the objective function.

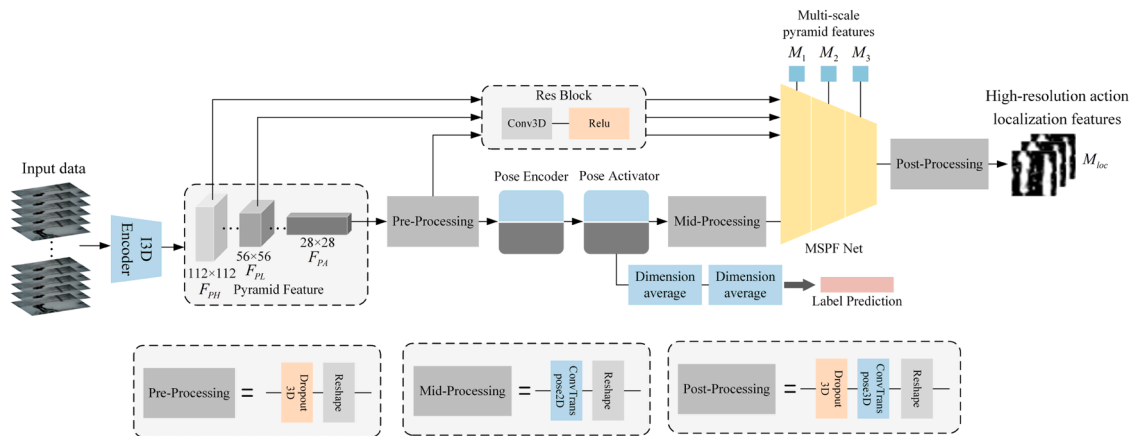


Fig. 1. Structural diagram of the proposed method. First, a video clip with 8 frames is input into the I3D backbone network to extract multi-scale primary features. Then, the primary features are fed into the Pose Encoder and Pose Activator for feature enhancement. Next, the enhanced features and residual features are input into the Multi-Scale Pyramid-Former Network for multi-scale feature fusion. Finally, post-processing is performed on the multi-scale fused features to obtain the model's prediction results.

3.3. Pose Encoder and Pose Activator

The basic features extracted solely by the backbone network have insufficient representational capacity and lack video context information. Therefore, this paper employs the Pose Encoder and Pose Activator to enhance features and capture crucial context information in videos. The network architecture of Pose Encoder and Pose Activator is shown in Fig. 2. Specifically, we input the primary low-resolution abstract semantic feature F_{PA} processed via pre-processing into two independent convolutional spaces to learn action domain information and activation domain information, respectively. After the activation domain, we add a sigmoid activation layer to generate an activation mask matrix M_A with values ranging from 0 to 1, which is used to activate the action localization feature F_P and label prediction. The modeling formula is as follows. Here, $Conv(\cdot)$ represents convolutional space, $Sigmoid(\cdot)$ represents sigmoid activation, P represents predicted label, and $Softmax(\cdot)$ represents the Softmax activation.

$$F_P = Conv(F_{PA}), M_A = Sigmoid(Conv(F_{PA})) \quad (1)$$

$$[F_P, M_A] = Conv([F_P, M_A]), [F_P, M_A] \in \mathbb{R}^{(B \cdot C) \times H \times W \times T} \quad (2)$$

$$P = \argmax(Softmax(M_A)) \quad (3)$$

Although the I3D backbone effectively captures action features with inherent spatiotemporal characteristics, it fails to incorporate essential global contextual details critical for comprehensive understanding. Therefore, we design an action temporal continuity encoder behind the action localization feature F_P of the Pose Activator, which is a small and lightweight Transformer. Specifically, the number of video frames T is 8, so we divide the dimensions of F_P equally into 4 parts and perform adjacent frames contextual encoding on them. We divide the dimension into four equal parts instead of directly processing the whole, because if the temporal span of global attention is too long, the model's discriminative ability is easily compromised by features with large temporal gaps and low correlation. Contextual encoding of temporal actions across adjacent frames not only maintains temporal continuity, but also prevents the network from focusing excessively on irrelevant information. We divide the weight matrix $Q/K/V$ in the Transformer into four equal dimensions which effectively reduces computational complexity, improves model efficiency, and prevents overfitting. Finally, we multiply the dimensionally concatenated F_P with the activation mask M_A to activate the action features. The modeling formula is as follows. Here, $LN(\cdot)$ represents layer normalization, $Dropout(\cdot)$ represents random dropout, $MSA(\cdot)$ represents multi-head self-attention, $MLP(\cdot)$ represents multi-layer perceptron layers, and $CNT(\cdot)$ represents dimensional concatenation, $\otimes$ denotes matrix element-wise multiplication.

$$F_P = [F_{P1}, F_{P2}, F_{P3}, F_{P4}], F_{Pi} \in \mathbb{R}^{(B \cdot C) \times H \times W \times 2}, i \in [1, 2, 3, 4] \quad (4)$$

$$F_{Pi} = LN(F_{Pi} + Dropout(MSA(F_{Pi}))), i \in [1, 2, 3, 4] \quad (5)$$

$$F_P = CNT(LN(F_{Pi} + MLP(F_{Pi}))), i \in [1, 2, 3, 4] \quad (6)$$

$$F_P = F_P \otimes M_A \quad (7)$$

3.4. Multi-Scale Pyramid-Former Network

In semi-supervised learning, extracting richer modal information from unlabeled data is crucial. The logic diagram of the proposed fusion method is illustrated in Fig. 3. We use the proposed Multi-Scale Pyramid-Former Network to perform the Pyramid Fusion Strategy on high-resolution global features, local detail features, and low-resolution abstract semantic features to obtain richer multi-scale information. This not only helps the model learn higher-dimensional modal information, but also allows us to use this rich multi-

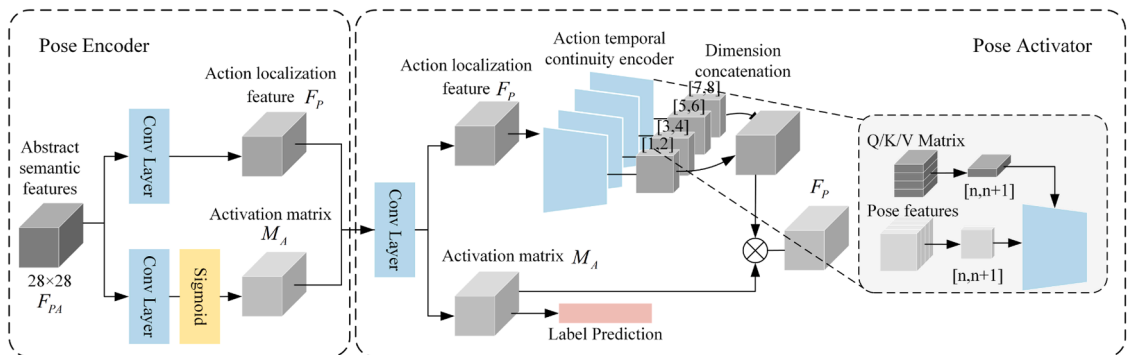


Fig. 2. Network architectures of the Pose Encoder and Pose Activator. This module accomplishes spatial mapping and dimensional transformation of features via convolutional layers, while leveraging an action temporal continuity encoder to perform contextual encoding of video features. In this figure, $\otimes$ denotes matrix element-wise multiplication.

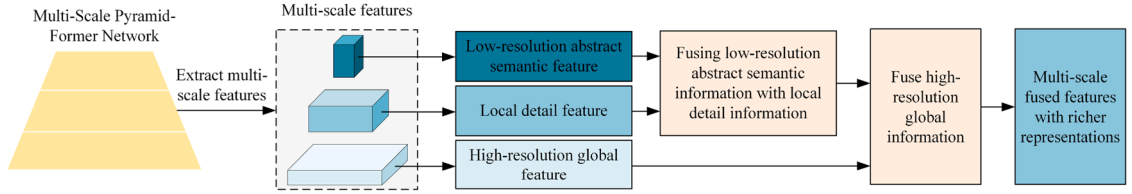


Fig. 3. Logic diagram of the proposed fusion method. We categorize multi-scale features into high-resolution global features, local detail features, and low-resolution abstract semantic features based on their dimensions, height, and width. First, we fuse the low-resolution abstract semantic features with the local detail features; then we fuse these combined features with the high-resolution global features to obtain multi-scale information with richer representations.

scale information for consistency regularization constraints in semi-supervised learning. This paper first introduces the implementation details of the MSPF Net in Section 3.4.1, and Section 3.4.2 details the Pyramid Fusion Strategy based on MSPF Net.

3.4.1. Network structure

To enhance the precision of action localization in our proposed framework, we continuously upsample the low-resolution semantic features with a size of 28×28 to the high-resolution features with a size of 224×224 . This operation not only enables the model to focus more on subtle changes in actions but also facilitates subsequent visualization analysis because high-resolution restored features are more interpretable than low-resolution abstract semantic features. The implementation details of the MSPF Net are shown in Fig. 4. Specifically, we first input the abstract semantic feature F_{28} of size 28×28 obtained from the residual block into the MSPF Net Layer1. Each layer outputs two parts: Att map and the multi-scale action localization feature $F_{M28/56/112}$. Att map is the attention map used for the Pyramid Fusion Strategy, and $F_{M28/56/112}$ is used to concatenate with the activated action localization feature F_p from the backbone network to fuse multi-scale information to obtain feature F_{M+p} . Both F_{M28} and F_p have a size of 28×28 . Second, the action localization feature F_{M+p} is refined through deconvolution space, expanding its size to 56×56 . Then, the local detail feature F_{56} of size 56×56 obtained from the residual block is input into the MSPF Net Layer2, and the output F_{M56} from MSPF Net layer2 is concatenated with the F_{M+p} of size 56×56 . Through deconvolution layers, a high-resolution global feature F_{M+p} of size 112×112 is obtained, which integrates local detail information. Next, the high-resolution global feature F_{112} with a size of 112×112 obtained from the residual block is input into the MSPF Net Layer3. The only difference is that the third layer requires not only the Att map of the second layer, but also the Att map of the first layer for pyramid fusion. Finally, we obtain a multi-scale action localization feature F_o that integrates high-resolution global features, local detail features, and low-resolution abstract semantic features, with a size of 224×224 . It is worth noting that we use the multi-scale action localization features $\{F_{M28}, F_{M56}, F_{M112}\}$ output by each layer of MSPF Net as the multi-scale pyramid features $\{M_1, M_2, M_3\}$ after passing through the ReLU activation layer for subsequent Coarse-to-Fine Multi-Scale Contrastive

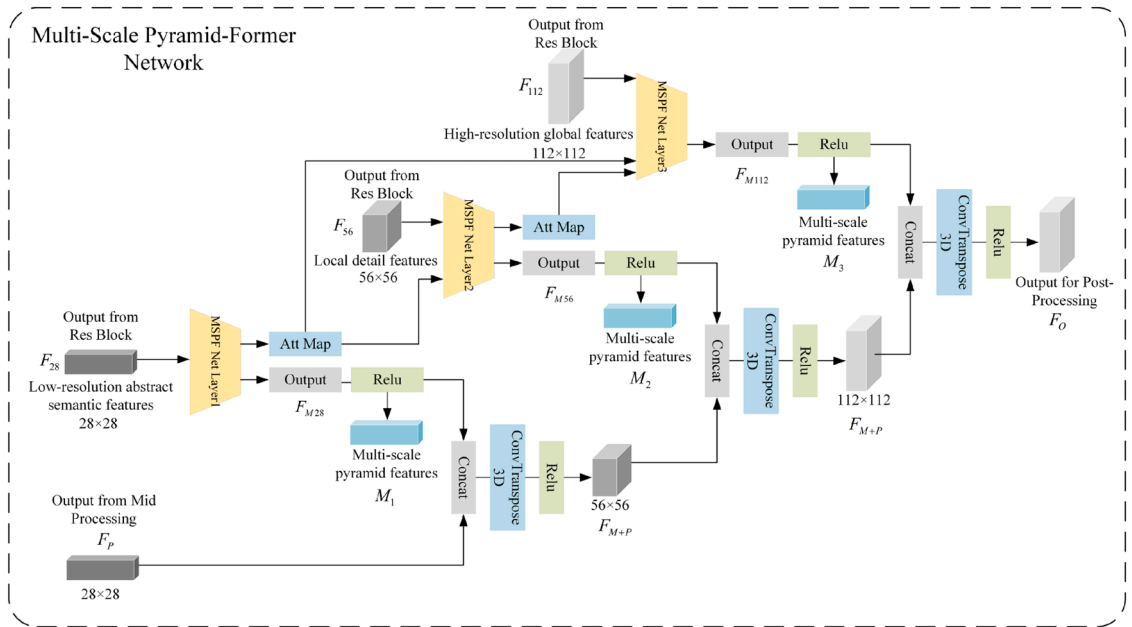


Fig. 4. Implementation details of the Multi-Scale Pyramid-Former Network. We reconstruct the 28×28 abstract semantic features into 224×224 high-resolution localization features using deconvolution.

Strategy.

3.4.2. Pyramid fusion strategy

As the number of convolutional layers increases, the output features contain increasingly more semantic information and have higher dimensions. Therefore, how to fuse multi-scale information in MSPF Net is of crucial importance. This paper proposes a Pyramid Fusion Strategy based on MSPF Net to integrate multi-scale information; details are shown in Fig. 5. Specifically, after inputting the multi-scale information obtained from the residual block into MSPF Net Layer1, Layer2, and Layer3, we obtain the corresponding query matrix Q , key matrix K , and value matrix V for each layer. The modeling process is illustrated below, where F_{28} denotes low-resolution abstract semantic features, F_{56} represents local detail features, F_{112} represents high-resolution global features, T represents matrix transpose, and W_Q , W_K , and W_V are learnable parameters in linear transformations.

$$\begin{bmatrix} Q_{28} & Q_{56} & Q_{112} \\ K_{28} & K_{56} & K_{112} \\ V_{28} & V_{56} & V_{112} \end{bmatrix} = \begin{bmatrix} W_Q \\ W_K \\ W_V \end{bmatrix} \times \begin{bmatrix} F_{28} \\ F_{56} \\ F_{112} \end{bmatrix}^T \quad (8)$$

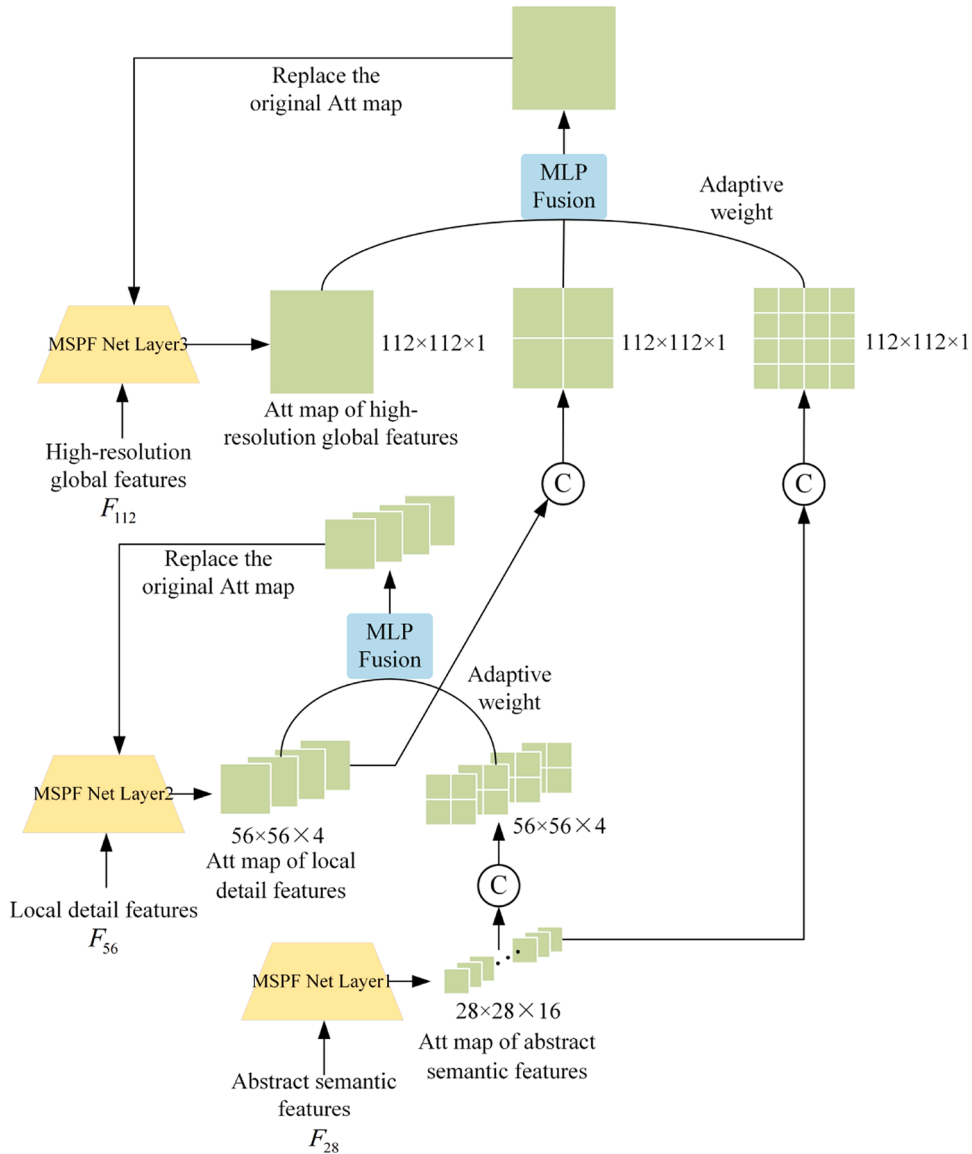


Fig. 5. Flowchart of the Pyramid Fusion Strategy. To achieve more effective multi-scale fusion, we concatenate the feature map with a size of $28 \times 28 \times 16$ into features with sizes of $56 \times 56 \times 4$ and $112 \times 112 \times 1$ through a convolutional mapping concatenation operation. In this figure, C represents the convolutional mapping concatenation operation.

The original Att map for each layer is obtained by multiplying the query matrix Q and key matrix K , followed by Softmax activation and random dropout. The process formula is as follows, where $\{A_{P28}, A_{P56}, A_{P112}\}$ represents the original Att map and T represents the number of video frames.

$$A_{P28} = Dropout\left(Softmax\left(\frac{Q_{28}K_{28}^T}{\sqrt{T}}\right)\right) \quad (9)$$

$$A_{P56} = Dropout\left(Softmax\left(\frac{Q_{56}K_{56}^T}{\sqrt{T}}\right)\right) \quad (10)$$

It is known that each element in the feature matrix output by the convolutional space contains feature information from the preceding convolutional space. As the convolution depth increases, the dimensions of the output features also increase, and the corresponding receptive field expands accordingly. Therefore, this paper performs a convolutional mapping concatenation operation on the Att map corresponding to the higher-dimensional abstract semantic features that have undergone deeper convolution in the MSPF Net Layer1. Specifically, we concatenate Att maps of size $28 \times 28 \times 16$ in the order of the convolutional kernel movement to obtain Att maps of size $56 \times 56 \times 4$ and $112 \times 112 \times 1$, respectively. Similarly, the Att map with a size of $56 \times 56 \times 4$ corresponding to the local detail features in the MSPF Net Layer2 is concatenated via convolutional mapping to obtain an Att map with a size of $112 \times 112 \times 1$. The process formula is as follows, where A_m^n represents the Att map of size $n \times n$ obtained by applying the convolutional mapping concatenation to the Att map of size $m \times m$, and $\Psi_m^n(\cdot)$ represents the convolutional mapping concatenation operation.

$$A_{28}^{56} = \Psi_{28}^{56}(A_{P28}), A_{28}^{112} = \Psi_{28}^{112}(A_{P28}), A_{56}^{112} = \Psi_{56}^{112}(A_{P56}) \quad (11)$$

Subsequently, we perform weighted fusion between the Att map processed by the convolutional mapping concatenation operation and the original Att map. Weighted fusion achieves adaptive learning of weight parameters through the parameter gradient optimization mechanism. Specifically, Att maps of different scales are fed into the fully connected layer, and the weight parameter $w = f_\theta(Att)$ is modeled using the trainable parameters in this layer. During the training phase, the weight w is updated synchronously with network parameters, enabling it to dynamically adapt to the optimal weight value based on training results. By replacing static weight values with adaptive updates of weight w , the network can dynamically adjust weight magnitudes to adapt to different tasks and enhance model robustness. The new Att map obtained after weighted fusion operation not only contains global information that the model needs to focus on at the current scale, but also includes (local/semantic) information that the model needs to focus on at other scales. Finally, we replace the initial Att map with the new Att map obtained, multiply it by the value matrix V , and obtain localization features $F_{M28/56/112}$ at different scales. The modeling process is as follows, where $WF(\cdot)$ represents weighted fusion and $MLP(\cdot)$ represents multi-layer perceptron layers.

$$A_{28} = A_{P28}, A_{56} = WF(A_{P56}, A_{28}^{56}), A_{112} = WF(A_{P112}, A_{28}^{112}, A_{56}^{112}) \quad (12)$$

$$F_{M28} = LN(F_{28} + Dropout(A_{28} \otimes V)) \quad (13)$$

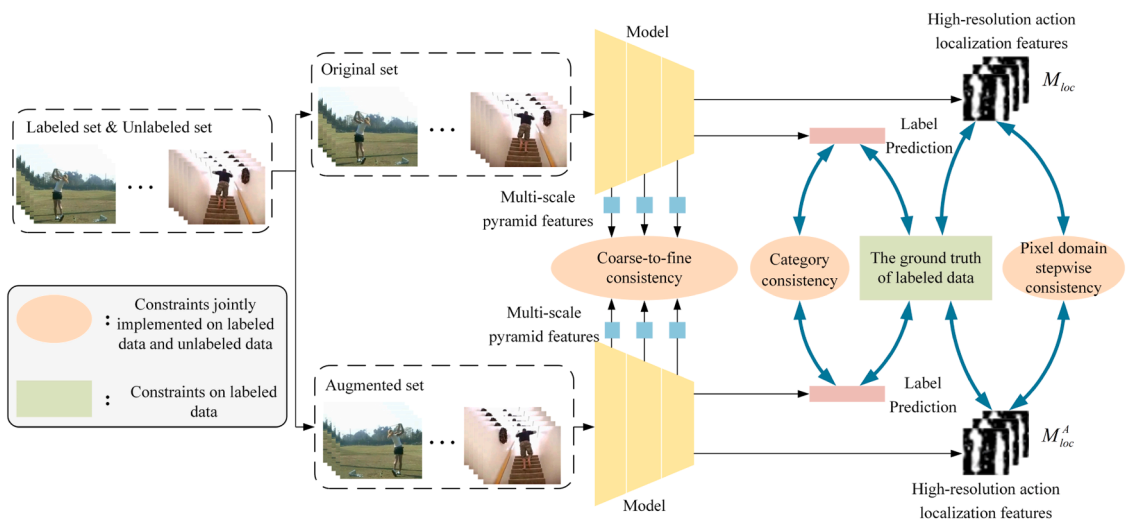


Fig. 6. Process of multiple consistency constraints based on MSPF Net. In this figure, squares represent consistency constraint applied on labeled data, and ellipses represent consistency constraints jointly applied to unlabeled data and labeled data. The multiple consistency constraints in this paper include the Coarse-to-Fine Multi-Scale Contrastive Strategy, the Pixel Domain Stepwise Localization Refinement Strategy, and Category Consistency.

$$F_{M28} = LN(F_{M28} + MLP(F_{M28})) \quad (14)$$

$$F_{M56} = LN(F_{56} + Dropout(A_{56} \otimes V)) \quad (15)$$

$$F_{M56} = LN(F_{M56} + MLP(F_{M56})) \quad (16)$$

$$F_{M112} = LN(F_{M112} + MLP(F_{M112})) \quad (17)$$

3.5. Semi-supervised multiple consistency constraints

In semi-supervised learning, adding certain perturbations to input data should result in consistent predictions from the model. This paper applies constraints to unlabeled data based on the consistency regularization. On the basis of MSPF Net, this paper proposes the Coarse-to-Fine Multi-Scale Contrastive Strategy and the Pixel Domain Stepwise Localization Refinement Strategy. The consistency constraints flowchart is shown in Fig. 6. Specifically, we apply data augmentation to the raw dataset and subsequently feeding both the initial and augmented datasets into two weight-sharing models. We then apply the Coarse-to-Fine Multi-Scale Contrastive Strategy to the multi-scale pyramid features output by MSPF Net, the Pixel Domain Stepwise Localization Refinement Strategy to the high-resolution action localization features, and a category consistency constraint to the classification results. Next, a detailed introduction to each strategy is provided below.

3.5.1. Coarse-to-fine multi-scale contrastive strategy

In order to enhance the robustness of feature representation for unlabeled data in MSPF Net, we apply the Coarse-to-Fine Multi-Scale Contrastive Strategy to the multi-scale pyramid features $\{M_1, M_2, M_3\}$ at different scales output by each layer of MSPF Net. The extraction method of multi-scale pyramid features $\{M_1, M_2, M_3\}$ has been detailed in Section 3.4.1. The reason it is called coarse-to-fine is that the resolution of the features corresponding to each layer of MSPF Net increases progressively. As the number of layers increases, the feature description of actions progresses from abstract to detailed. Specifically, since multi-scale pyramidal features $\{M_1, M_2, M_3\}$ (from the original data) and $\{M_1^A, M_2^A, M_3^A\}$ (from the augmented data) derive from the same action instance, their discriminative properties should be consistent. Therefore, we treat the original features and augmented features $\{M_1, M_1^A\}$, $\{M_2, M_2^A\}$ and $\{M_3, M_3^A\}$ as positive sample pairs, and measure the similarity between original features and augmented features within the positive sample pairs at different scales. It is noteworthy that, due to changes in action positions after data augmentation, the actions localization features will also shift corresponding to the augmentation method. Therefore, we need to perform inverse augmentation on the localization features of augmented data. For example, when using horizontal flipping as a data augmentation method, the feature maps corresponding to the augmented data should be horizontally flipped back before applying the Coarse-to-Fine Multi-Scale Contrastive Strategy. The Coarse-to-Fine Multi-Scale Contrastive Strategy is calculated as follows:

$$C2F(X, Y) = -\frac{1}{B \times C} \sum_{i=1}^B \sum_{j=1}^C \log \frac{\exp\left(\frac{1}{1+DIST_{ij}(X, Y)}\right)}{\sum_{m \neq i}^B \sum_{n \neq j}^C \exp\left(\frac{1}{1+DIST_{mn}(X, Y)}\right)} \quad (19)$$

Among them, $DIST(\cdot)$ is used to measure the distance between features, and the calculation formula is described as follows:

$$DIST(X, Y) = \left(\sum_{i=1}^H \sum_{j=1}^W |X_{ij} - Y_{ij}|^2 \right)^{\frac{1}{2}}, X \& Y \in \mathbb{R}^{B \times C \times H \times W}, DIST(\cdot) \in \mathbb{R}^{B \times C} \quad (20)$$

After applying the Coarse-to-Fine Multi-Scale Contrastive Strategy to the multi-scale pyramid features at each layer of MSPF Net, we perform a weighted summation. The overall calculation formula is shown below, where $A(\cdot)^{-1}$ denotes the inverse augmentation transformation.

$$L_{C2F} = \frac{1}{3} \left[C2F(M_1, A(M_1^A)^{-1}) + C2F(M_2, A(M_2^A)^{-1}) + C2F(M_3, A(M_3^A)^{-1}) \right] \quad (21)$$

3.5.2. Pixel domain stepwise localization refinement strategy

Due to the presence of dominant backgrounds and static regions in videos, the model may enhance the feature representation of some backgrounds while weakening action foreground representation. Therefore, in order to refine action detection boundaries and enhance the model's focus on the salient aspects of action sequences, this paper proposes the Pixel Domain Stepwise Localization Refinement Strategy. Specifically, we apply pixel domain consistency regularization constraints to the high-resolution action localization features output by the model with a size of 224×224 . Similar to the Coarse-to-Fine Multi-Scale Contrastive Strategy, we first perform inverse augmentation on the action localization feature M_{loc}^A obtained from augmented data to ensure the same mapping relationship. For both the augmented and original data, the model should predict action positions consistently. Therefore, we measure the difference between M_{loc} and M_{loc}^A using Mean Square Error (MSE) loss. The calculation process is described as follows, where $MSE(\cdot)$ represents the MSE loss.

$$L_{loc} = MSE\left(M_{loc}, A\langle M_{loc}^A \rangle^{-1}\right) \quad (22)$$

Actions in videos exhibit continuity in the temporal dimension, meaning that pixels in the action foreground area of adjacent frames move to neighboring regions due to the action motion, while background pixels remain stationary or change minimally. Therefore, the corresponding action localization features should also possess continuity. To capture this continuity, we impose consistency constraints on the adjacent-frame migration maps of the initial and the augmented dataset. Specifically, we calculate the deviation between the action localization feature of adjacent frames to obtain the adjacent-frame action migration maps. This deviation has a relatively large value in the action foreground area, but is relatively smaller in the background and static regions. Through this method, the model can effectively capture the movement trend of actions. In video action detection, the detection of action foregrounds is often affected by dominant background interference. To suppress this negative impact, we design a threshold activation neighboring frame action migration constraint. Specifically, in the adjacent-frame action migration map, the weights of irrelevant background and static regions are relatively small. We eliminate the weights of the background regions and increase the weights of the action migration regions, thereby facilitating the model's concentrated attention on action regions. The threshold activation neighboring frame action migration constraint is described as follows. Here, $Mask_n$ represents the threshold mask corresponding to the n th frame, M_{loc}^n represents the localization feature corresponding to the n th frame, $\otimes$ represents the matrix element-wise multiplication, M_F represents the adjacent-frame action migration map corresponding to the original data, and M_F^A represents the adjacent-frame action migration map corresponding to the augmented data.

$$Mask_n = \begin{cases} 3/2, & \text{if } M_{loc}^n(i, j) - M_{loc}^{n-1}(i, j) > Avg \\ 0, & \text{otherwise} \end{cases}, \text{ where } Avg = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W M_{loc}^n(i, j) \quad (23)$$

$$M_n^F = (M_{loc}^n - M_{loc}^{n-1})^2 \otimes Mask_n \quad (24)$$

$$M_F = [M_2^F, M_3^F, \dots, M_n^F] \quad (25)$$

$$L_f = MSE\left(M_F, A\langle M_F^A \rangle^{-1}\right) \quad (26)$$

We obtained the adjacent-frame action migration map M_F from the above equation. M_F can effectively capture the migration trend of actions. The background area is suppressed after passing through the threshold mask, while the action migration area remains at a higher value. In order to better refine action detection boundaries, we employ a gradient mechanism to enhance the model's attention to action regions. On the one hand, actions in videos exhibit temporal continuity. The migration of pixels in the foreground regions of actions in adjacent frames, resulting from motion, exhibits smoothness and directionality. The directionality and continuity of pixels can be quantified using mathematical derivative operations. First-order derivatives can effectively characterize the directionality of pixel changes and the intensity of these changes, while second-order derivatives are more sensitive to abrupt changes in the rate of pixel changes. On the other hand, background regions differ from action foregrounds because they are typically stationary or undergo only weak changes. Consequently, their pixel change rate is far lower than that of action regions. Capturing the intensity of pixel changes using the first-order derivative can effectively distinguish the action foreground from the video background, while the second-order derivative further captures action trends through changes in foreground pixel changes. We apply consistency constraints to the derivative results, and the calculation is shown below, where $dist$ denotes the distance between adjacent frames, M_{grad} represents the derivative result corresponding to the original data, and M_{grad}^A represents the derivative result corresponding to the augmented data.

$$M_{grad} = \frac{d^2(M_F)}{d(dist)^2} \quad (27)$$

$$L_g = MSE\left(M_{grad}, A\langle M_{grad}^A \rangle^{-1}\right) \quad (28)$$

Finally, the Pixel Domain Stepwise Localization Refinement Strategy is described as follows.

$$L_{pixel} = L_{loc} + L_f + L_g \quad (29)$$

3.5.3. Category consistency and supervised loss

Finally, we apply consistency regularization constraint to the category predictions of the initial dataset and augmented dataset. The category consistency is described as follows, where P represents the model's category prediction for the original data, P_A represents the model's category prediction for the augmented data, and n_Class represents the number of action categories.

$$KL(P \parallel P_A) = \sum P \cdot \log \frac{P}{P_A}, P \& P_A \in \mathbb{R}^{B \times n_Class} \quad (30)$$

$$L_C = \frac{1}{2} KL\left(P \parallel \frac{P + P_A}{2}\right) + \frac{1}{2} KL\left(P_A \parallel \frac{P + P_A}{2}\right) \quad (31)$$

For supervised loss, we implement it through simple BCE loss and Dice loss, as shown below, where y_i represents the ground truth category label and $P(y_i)$ represents the model's prediction of the category, $y_i, P(y_i) \in \mathbb{R}^{B \times n_{Class}}$, X_i represents the ground truth label for action localization, $P(X_i)$ represents the model's prediction of localization, $X_i, P(X_i) \in \mathbb{R}^{B \times C \times T \times 224 \times 224}$, N represents the number of labeled samples.

$$L_{label} = -\frac{1}{N} \left\{ \sum_{i=1}^N \left[y_i \cdot \log(P(y_i)) + (1 - y_i) \cdot \log(1 - P(y_i)) + \left(1 - \frac{2|X_i \cap P(X_i)|}{|X_i| + |P(X_i)|} \right) \right] \right\} \quad (32)$$

The overall objective function of the model is described as follows, where w_{label} , w_C , w_{C2F} and w_{pixel} are the weight hyperparameters.

$$L_{total} = w_{label}L_{label} + w_C L_C + w_{C2F} L_{C2F} + w_{pixel} L_{pixel} \quad (33)$$

4. Experiments

4.1. Datasets

To validate the efficacy of MSPF Net, we perform comprehensive experiments on two widely recognized benchmark datasets: JHMDB-21 and UCF101-24. Dataset information is compared in Table 1.

JHMDB-21 (Jhuang et al., 2013) comprises 928 video clips across 21 distinct action categories, mostly belonging to the motion categories. Each category has 36-55 video samples, and the original resolution of each sample data is 320×240 , each video contains 14-40 frames, with a total of 31383 images annotated on the JHMDB-21 dataset.

UCF101-24 (Soomro et al., 2012) comprises 3207 video clips, and the original resolution of the video samples is 320×240 , each video contains approximately 150 frames and 24 action classes, with 2284 training videos and 923 testing videos.

4.2. Implementation details

In this study, we set the input frames resolution to 224×224 to facilitate pyramid fusion, and set the number of video frames to 8. The batch size in the experiment is set to 8, and both labeled and unlabeled samples are mixed and fed into the model for training. The model is optimized using the Adam optimizer, with an initial learning rate of $1e-4$. A learning rate decay mechanism is implemented, triggered when the loss function fails to decrease for 5 consecutive epochs, the learning rate is reduced to 50 % of its current value. For the weight hyperparameters, we set w_{label} and w_C to 1, w_{C2F} to 0.7, and w_{pixel} to 0.3. Regarding the number of training epochs, we train the model for 70 epochs on JHMDB-21 dataset, with convergence achieved at approximately 50 epochs; for the UCF101-24 dataset, we train the model for 150 epochs, with convergence at approximately 120 epochs. Three independent experiments are performed using random seeds, and the mean value is calculated as the final result. The experiments are conducted utilizing the PyTorch framework, with computational tasks accelerated by a GeForce RTX 3090 GPU.

4.3. Comparison with state-of-the-art semi-supervised methods

In the experiments on JHMDB-21 and UCF101-24, we evaluate the action detection performance using two evaluation metrics: frame-level mean average precision (f-mAP) and video-level mean average precision (v-mAP). For the JHMDB-21 dataset, the data annotation rate is 30 %. The number of labeled data used by other advanced semi-supervised methods is consistent with ours. In subsequent experiments on JHMDB-21, unless otherwise specified, we evaluate the semi-supervised performance of the model with a 30 % labeling ratio. The literature comparisons of MSPF Net with other state-of-the-art baseline methods on JHMDB-21 are presented in Table 2, where the experimental results are obtained from the original literature.

The experimental results from JHMDB-21 indicate that when using only 30 % of the labeled data, the proposed MSPF Net achieves competitive detection accuracy. Compared with the state-of-the-art method Multi-FB (Zhang et al., 2025), our proposed method outperforms Multi-FB by 3.5 % in terms of f-mAP@0.5, by 0.7 % in terms of v-mAP@0.2, and by 3.1 % in terms of v-mAP@0.5.

UCF101-24 is a large dataset. In this study, only 20 % of the dataset is used as labeled data, with the remaining 80 % used as unlabeled data. The number of labeled data used by other advanced semi-supervised methods is consistent with ours. In subsequent experiments on UCF101-24, unless otherwise specified, we evaluate the semi-supervised performance of the model with a 20 % labeling ratio. The literature comparisons of MSPF Net with other state-of-the-art baseline methods on UCF101-24 are presented in Table 3, where the experimental results are obtained from the original literature.

Experimental results on UCF101-24 and JHMDB-21 suggest that:

Table 1

Comparison results of relevant information between the JHMDB-21 and the UCF101-24 datasets, where $\sim$ denotes approximate values.

Datasets	Number of categories	Number of videos	Number of frames in each video	Video resolution
UCF101-24 (Soomro et al., 2012)	24	3207	~ 150	320×240
JHMDB-21 (Jhuang et al., 2013)	21	928	14-40	320×240

Table 2

Comparison with state-of-the-art methods on the JHMDB-21.

Methods	Backbone	JHMDB-21		
		3-D	f-mAP	v-mAP
			0.5	0.2
MixMatch (Berthelot et al., 2019)	✓		7.5	46.2
GLNet (Zhang et al., 2020)	✓		65.9	77.3
Pseudo-label (Lee, 2013)	✓		57.4	90.1
Co-SSD(CC) (Jeong et al., 2019)	✓		60.7	94.3
ETEVAD (Kumar & Rawat, 2022)	✓		64.4	95.4
BWCC (Zhong et al., 2023)	✓		66.3	94.8
M-T SSL (Singh et al., 2024)	✓		68.5	98.4
Multi-FB (Zhang et al., 2025)	✓		69.2	95.7
MSPF Net (ours)	✓		71.6	96.4

Table 3

Comparison with state-of-the-art methods on UCF101-24.

Methods	Backbone		UCF101-24		
	2-D	3-D	f-mAP	v-mAP	
	S		0.5	0.2	0.5
LAVP (Mettes et al., 2017)	✓		-	37.4	-
PSAL (Mettes & Snoek, 2018)	✓		-	41.8	-
FMAL (Chéron et al., 2018)		✓	-	43.9	17.7
ASST (Escorcía et al., 2020)		✓	45.8	19.3	-
UWAD (Arnab et al., 2020)		✓	-	61.7	35.0
MixMatch (Berthelot et al., 2019)		✓	20.2	60.2	13.8
GLNet (Zhang et al., 2020)		✓	30.4	45.5	17.3
Pseudo-label (Lee, 2013)		✓	64.9	93.0	65.6
Co-SSD(CC) (Jeong et al., 2019)		✓	65.3	93.7	67.5
ETEVAD (Kumar & Rawat, 2022)		✓	69.9	95.7	72.1
BWCC (Zhong et al., 2023)		✓	71.9	95.3	73.7
M-T SSL (Singh et al., 2024)		✓	69.8	94.9	72.2
Multi-FB (Zhang et al., 2025)		✓	72.8	96.1	73.9
MSPF Net (ours)		✓	73.3	96.7	74.6

- (1) On the UCF101-24 dataset, when using only 20 % of the labeled data, the proposed method still demonstrates an advantage in both f-mAP and v-mAP metrics. Compared to the state-of-the-art method Multi-FB (Zhang et al., 2025), MSPF Net outperforms Multi-FB by 0.7 % on f-mAP@0.5 metric, by 0.6 % on v-mAP@0.2, and by 0.9 % on v-mAP@0.5.
- (2) The MSPF Net introduced in this study exhibits performance on par with cutting-edge approaches across both the JHMDB-21 and UCF101-24 datasets. This not only validates the efficacy of MSPF Net but also demonstrates its strong robustness and scalability. The experimental results indicate that MSPF Net is capable of extracting richer scale information even in the absence of annotated data.

4.4. Further analysis

4.4.1. Comparison with fully supervised methods

The proposed method exhibits significant superiority under semi-supervised learning scenarios. In order to explore the performance of MSPF Net when using all labeled samples, this section conducts experiments based on JHMDB-21. We gradually increase the

Table 4

Experimental results of gradually increasing the proportion of labeled samples on JHMDB-21.

Labels	JHMDB-21			
	f-mAP		v-mAP	
	0.2	0.5	0.2	0.5
30 %	94.1	71.6	96.4	72.5
40 %	94.4	73.7	96.7	73.8
50 %	94.7	74.5	96.9	74.1
60 %	95.1	75.6	97.0	74.4
70 %	95.4	75.9	97.0	74.7
80 %	95.5	76.7	97.1	75.3
90 %	95.6	77.2	97.2	76.6
100 %	95.6	77.5	97.2	77.1

proportion of labeled samples starting from 30 % and evaluate the proposed approach against current leading fully supervised methods. Unlike semi-supervised cases, when using labels for all data, we apply supervised constraints to all data while also applying multiple consistency constraints. The results of the literature comparison with other methods are illustrated in Tables 4 and 5, where the experimental results are obtained from the original literature.

Regarding the experimental findings, this study presents the following perspectives:

- (1) As the proportion of labeled data gradually increases, it can be observed that MSPF Net's capability in detecting targets is gradually improving across the metrics f-mAP@0.2/0.5 and v-mAP@0.2/0.5, which demonstrates that MSPF Net can make efficient use of annotated data and the model design is reasonable and feasible.
- (2) On different evaluation metrics, when using 100 % labeled samples, the final performance improvement is achieved (1.6 %, 8.2 %, 0.8 %, 6.3 %). It is worth noting that the performance improvement at f-mAP@0.5 and v-mAP@0.5 is much greater than at f-mAP@0.2 and v-mAP@0.2. This is because as the threshold increases, there is a higher demand for the precision of action localization. MSPF Net can focus on the details of actions by integrating information from different scales. Additionally, the Pixel Domain Stepwise Localization Refinement Strategy can better refine the boundaries, resulting in greater improvements in high-threshold situations.
- (3) Compared to fully supervised methods, MSPF Net demonstrates performance on par with state-of-the-art techniques, particularly excelling in accuracy for the v-mAP@0.2 metric. This indicates that MSPF Net is not only suitable for semi-supervised learning scenarios but also retains significant advantages in fully supervised learning settings.
- (4) In the evaluation metric of v-mAP@0.5, the performance of MSPF Net is lower than that of the method of AL-2D-3D (Tong et al., 2022), because Tong et al. not only use a 3D backbone network but also a 2D backbone network, which results in a relatively larger model size and longer computation time.

4.4.2. Comparison with other multi-scale pyramid network structures

To demonstrate the advantages of the proposed MSPF Net under semi-supervised conditions, we re-implement representative multi-scale pyramid network models under the same experimental settings and compare them with our method. It is important to highlight that the SSV (Xiao et al., 2022) method requires forward and backward action information, so we use the forward and backward difference values of adjacent video frames as model's inputs. Furthermore, we combine the pyramid features output by PiT (Zhang et al., 2022) with the decoder of U-Net (Ronneberger et al., 2015) to better adapt to the detection task. The experimental results are presented in Table 6, where * represents re-implementation, and "Inference" represents the time required for the model to process an 8-frame video segment.

The experimental findings presented in this study lead to the following conclusions:

- (1) The methods of U-Net, SSV, and PiT exhibit low detection accuracy under semi-supervised conditions. This is because multi-scale pyramid networks demonstrate lower robustness due to the influence of unlabeled data. This paper enhances the similarity of feature representations at each layer of the pyramid model by applying the Coarse-to-Fine Multi-Scale Contrastive Strategy to MSPF Net, thereby effectively improving the robustness of model predictions.
- (2) Although U-Net, SSV, and PiT all utilize multi-scale information, they only employ a simple dimensional concatenation method for fusion. In contrast, we utilize a Pyramid Fusion Strategy to merge and replace the Att maps output by different layers of MSPF Net, enabling the model to exhibit better attention to information at different scales.

Table 5
Comparison with state-of-the-art fully supervised learning methods.

Fully supervised methods	Backbone		JHMDB-21		
	2-D	3-D	f-mAP	v-mAP	
TACNet (Sogn et al., 2019)	✓		0.5	0.2	0.5
DFNet (Zhao & Snoek, 2019)	✓		65.5	74.1	73.4
MOC (Li et al., 2020)	✓		-	-	74.7
SAMOC (Ma et al., 2021)	✓		70.8	77.3	70.2
SparseDense (Li et al., 2020)		✓	73.1	79.4	71.3
T-CNN (Hou et al., 2017)		✓	-	76.1	74.3
Videocapsulenet (Duarte et al., 2018)		✓	61.3	78.4	76.9
ETEAD (Kumar & Rawat, 2022)		✓	64.6	95.1	-
P3D-CTN (Wei et al., 2019)	✓	✓	68.1	96.8	68.4
AL-2D-3D (Tong et al., 2022)	✓	✓	71.1	84.0	80.5
CSTA-Net (Yang et al., 2023)	✓	✓	76.3	90.7	90.0
YOWOV3 (Zhu et al., 2024)		✓	73.4	79.4	79.5
YOWOV3-TSM (Zhu et al., 2024)		✓	63.0	-	-
STAD-2D-3D (Liu et al., 2024)	✓	✓	60.3	-	-
Multi-FB (Zhang et al., 2025)		✓	73.5	-	-
MSPF Net (ours)		✓	77.2	96.1	76.8
			77.5	97.2	77.1

Table 6

The comparison results of MSPF Net with other multi-scale pyramid networks on the JHMDB-21 dataset, where * represents re-implementation.

Methods	Labels	GFLOPs	Parameters	f-mAP		v-mAP		Inference (in secs)
				0.2	0.5	0.2	0.5	
U-Net* (Ronneberger et al., 2015)	30 %	139.328G	31.042M	87.6	58.3	90.7	55.6	0.901
U-Net* (Ronneberger et al., 2015) + Attention	30 %	189.611G	42.357M	90.9	61.2	93.3	60.7	1.175
SSV* (Xiao et al., 2022)	30 %	443.833G	57.176M	91.3	64.6	93.7	65.0	1.913
PiT* (Zhang et al., 2022)	30 %	284.508G	130.112M	93.6	67.0	95.1	68.3	2.572
MSPF Net (Ours)	30 %	231.860G	48.435M	94.1	71.6	96.4	72.5	0.932

- (3) Compared to the methods of SSV and PiT, our method has fewer GFLOPs and model parameters, as well as a shorter inference time. This is because SSV requires two data modalities as input, and PiT's method needs separate training for each generated multi-scale feature; thus, our method demonstrates lower time complexity. Compared to U-Net, although our method has slightly more model parameters, its detection accuracy is significantly higher than that of U-Net. Therefore, the proposed MSPF Net is more competitive.

4.4.3. Comparison with other contrastive learning methods

In this section, we further investigate the Coarse-to-Fine Multi-Scale Contrastive Strategy. We replace the Coarse-to-Fine Multi-Scale Contrastive Strategy with other advanced contrastive learning methods and re-implement the experiments under the same experimental conditions. Specifically, because the methods of ASVRL (Assefa et al., 2023) and TimeBalance (Dave et al., 2023) conduct contrastive learning not from a multi-scale perspective but from the perspective of different time segments within the same video, we combine the features from the last layer output of MSPF Net with the contrastive learning methods of ASVRL and TimeBalance. Table 7 presents the experimental results of this study, * represents re-implementation, where “C2F” represents the Coarse-to-Fine Multi-Scale Contrastive Strategy, and “Baseline” denotes the condition without the Coarse-to-Fine Multi-Scale Contrastive Strategy.

Experimental findings indicate that, although the methods developed by Assefa et al. and TimeBalance both utilize contrastive learning, their approaches are based on the perspective of temporal invariance. While these methods improve the detection accuracy of the model to some extent, they lack an understanding of information at different scales, resulting in limited capability to handle multi-scale features. In contrast, the Coarse-to-Fine Multi-Scale Contrastive Strategy improves the model's capacity to capture hierarchical information at different scales by comparing original features of various scales from different layers of MSPF Net with augmented features. Furthermore, by comparing features at different levels, the model better able to understand and capture the details and contextual information within videos, effectively enhancing its performance on the f-mAP and v-mAP metrics, thereby demonstrating its strong competitiveness and effectiveness in handling complex tasks.

4.4.4. Ablation experiments

4.4.4.1. The contributions of different modules in MSPF Net. We perform additional ablation studies to further investigate the different components of the Multi-Scale Pyramid-Former Network, including the MSPF Net, Coarse-to-Fine Multi-Scale Contrastive Strategy, and Pyramid Fusion Strategy. Specifically, when we remove MSPF Net, we connect the multi-scale residual information after the Pose Activator. The purpose of this processing is to compare the ability of the MSPF Net to fuse multi-scale information. When the Pyramid Fusion Strategy is not utilized, we integrate multi-scale information using reshape and addition. The experimental results are shown in Table 8, where “MSPF” denotes the Multi-Scale Pyramid-Former Network, “C2F” represents the Coarse-to-Fine Multi-Scale Contrastive Strategy, “PF” represents the Pyramid Fusion Strategy, “×” indicates non-utilization, and “Baseline” represents the non-use of all modules of the Multi-Scale Pyramid-Former Network. According to the experimental results, when all modules of the Multi-Scale Pyramid-Former Network are not utilized (Baseline), the model cannot effectively learn information at different scales, which leads to a poor action feature representation. After adding the MSPF Net, the model gains the ability to capture information at different scales, resulting in a significant increase in model performance. At the same time, both the Pyramid Fusion Strategy and the Coarse-to-Fine Multi-Scale Contrastive Strategy can further enhance the model's ability to capture hierarchical information at different scales. Ultimately, after integrating all modules together, the model achieves optimal performance.

Table 7

The experimental results comparing the Coarse-to-Fine Multi-Scale Contrastive Strategy with other contrastive learning methods, where * represents re-implementation.

Methods	JHMDB-21				UCF101-24			
	f-mAP		v-mAP		f-mAP		v-mAP	
	0.2	0.5	0.2	0.5	0.2	0.5	0.2	0.5
Baseline	93.1	68.3	95.0	69.2	93.7	71.7	95.8	72.9
ASVRL* (Assefa et al., 2023)	93.7	69.5	95.2	70.4	94.1	71.9	96.1	73.3
TimeBalance* (Dave et al., 2023)	93.8	70.3	95.9	71.7	94.3	72.5	96.3	73.8
C2F (Ours)	94.1	71.6	96.4	72.5	94.8	73.3	96.7	74.6

Table 8

The ablation experimental results of different modules in MSPF Net.

MSPF	PF	C2F	JHMDB-21				UCF101-24			
			f-mAP		v-mAP		f-mAP		v-mAP	
			0.2	0.5	0.2	0.5	0.2	0.5	0.2	0.5
Baseline			89.5	64.0	92.7	64.1	89.7	66.2	93.6	67.1
✓	×	×	92.6	67.5	94.8	68.4	93.2	71.0	95.6	71.2
✓	✓	×	93.1	68.3	95.0	69.2	93.7	71.7	95.8	72.9
✓	×	✓	93.4	68.5	95.2	69.7	94.1	72.5	96.2	73.1
✓	✓	✓	94.1	71.6	96.4	72.5	94.8	73.3	96.7	74.6

4.4.4.2. The contributions of multiple consistency constraints. This paper proposes a multiple consistency constraint method based on semi-supervised learning, which includes the Coarse-to-Fine Multi-Scale Contrastive Strategy, Pixel Domain Stepwise Localization Refinement Strategy, and category consistency constraint. The Pixel Domain Stepwise Localization Refinement Strategy is divided into action localization consistency L_{loc} , threshold activation neighboring frame action migration constraint L_f , and gradient consistency L_g . Table 8 demonstrates the effectiveness of the Coarse-to-Fine Multi-Scale Contrastive Strategy. Therefore, we conduct ablation studies to examine the contributions of the Pixel Domain Stepwise Localization Refinement Strategy and the category consistency constraint. The experimental results are shown in Table 9, where “P-D-S” represents the Pixel Domain Stepwise Localization Refinement Strategy, “Category” represents the category consistency constraint, and “Baseline” represents the scenario no constraints added.

The experimental findings reveal that on both datasets, compared to the baseline, adding only the category consistency constraint or only Pixel Domain Stepwise Localization Refinement Strategy significantly improves model performance. Meanwhile, when category consistency constraints are added, each component of the Pixel Domain Stepwise Localization Refinement Strategy also contributes positively to model performance. Obtaining the same conclusion on both datasets confirms the effectiveness and universality of the proposed multiple consistency constraints.

4.4.4.3. Visualization of contributions from each component. In this section, we visualize the magnitude of performance improvements contributed by each component as bar charts to more intuitively illustrate the impact of different components on overall performance. Specifically, we calculate the performance change value after adding a given component as the magnitude of performance improvement, as shown in Fig. 7. From the visualization results on the JHMDB-21 and UCF101-24 datasets, it can be observed that the Pyramid Fusion Strategy, Coarse-to-fine Multi-scale Contrastive Strategy, and Pixel Domain Stepwise Localization Refinement Strategy all significantly improve the model's performance on both datasets. Additionally, Category consistency also contributes to improving the model's accuracy.

4.4.5. Visualization and analysis

4.4.5.1. Visualization of MSPF Net. The Multi-Scale Pyramid-Former Network, as a key component of this study, is responsible for integrating high-resolution global features, local detail features, and low-resolution abstract semantic features, and for outputting the most representative discriminative features. To demonstrate how MSPF Net attends to action regions in videos, we visualize the attention map of the last layer in MSPF Net. The output is a heat map, which is stacked on the corresponding video frames. The visualization of MSPF Net is compared with those of state-of-the-art foreground detection methods, including Contextual Information Separation (CIS) (Yang et al., 2019), Unsupervised Foreground Attention (UFA) (Shi et al., 2021), and Multi-scale Spatiotemporal Feature-aware Action Detection (MSAD) (Miao & Ke, 2022). The visualization results of other methods are obtained from the original literature, as shown in Fig. 8.

From the visualization results, it can be seen that although CIS, UFA, and MSAD can focus on the action regions to some extent, their attention spans far exceed the actual action boundaries. Their regions of interest not only include excessive background areas but also show incomplete attention to the action foreground. In contrast, the proposed MSPF Net accurately focuses on the action regions without allocating unnecessary attention to the background areas. Furthermore, it avoids incomplete attention to the action foreground. This is achieved through the Pyramid Fusion Strategy, which integrates features from different scales and resolutions, making the model more sensitive to action details in unlabeled data.

In order to gain a more comprehensive understanding of the action information focused on by different layers in MSPF Net, we use “Golf” as input to visually analyze the attention maps of the first, second, and third layers in MSPF Net after the Pyramid Fusion Strategy, as shown in Fig. 9.

From the visualization results, it can be observed that the attention map of the first layer contains abstract semantic information, focusing more on small blocks and has lower interpretability. The second layer focuses more on local details, including some linear regions, while the third layer pays more attention to global information. For better interpretability, we extract attention maps from different channels in the third layer for visualization. From the 1st, 9th and 41st channels, it can be observed that the model already attends well to the action area with relatively high accuracy. It is worth noting that the attention points of the 7th, 52nd, and 66th channels are not located at the center of the actions, but rather at the action boundaries. This is because in the Pixel Domain Stepwise Localization Refinement Strategy, we compute derivatives for action regions, which makes the model focus more on action boundaries.

Table 9
Experimental results of ablation study on multiple consistency constraints.

Category	P-D-S	JHMDB-21				UCF101-24			
		f-mAP		v-mAP		f-mAP		v-mAP	
		0.2	0.5	0.2	0.5	0.2	0.5	0.2	0.5
Baseline		91.1	67.1	93.3	67.7	92.2	68.1	95.0	68.3
✓	×	91.8	68.5	94.2	69.3	92.5	70.4	95.4	70.8
×	✓	93.8	70.8	95.7	71.4	94.1	71.9	96.2	73.1
Category	L_{loc}	L_f	L_g	f-mAP	v-mAP	f-mAP	v-mAP	f-mAP	v-mAP
				0.2	0.5	0.2	0.5	0.2	0.5
Baseline				91.1	67.1	93.3	67.7	92.2	68.1
✓	✓	×	×	92.3	67.9	95.4	68.5	92.8	71.0
✓	✓	✓	×	93.0	69.0	95.8	69.5	93.1	72.4
✓	✓	✓	✓	94.1	71.6	96.4	72.5	94.8	73.3
								96.7	74.6

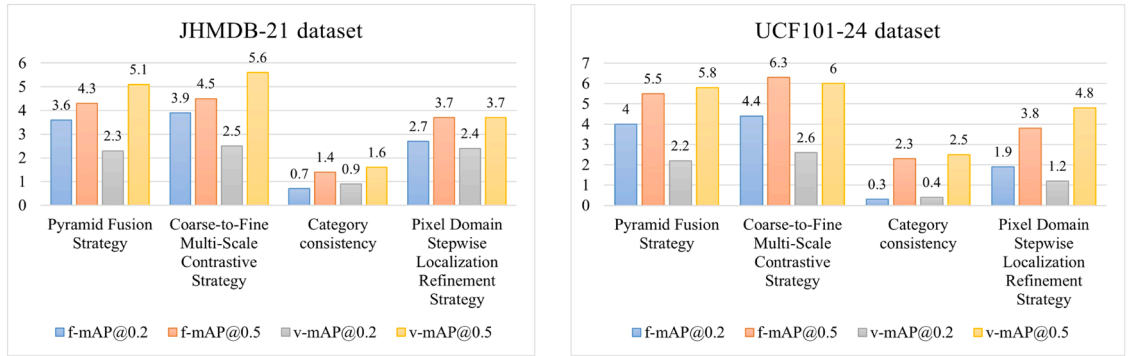


Fig. 7. Contributions of different components to model performance on the JHMDB-21 and UCF101-24 datasets.

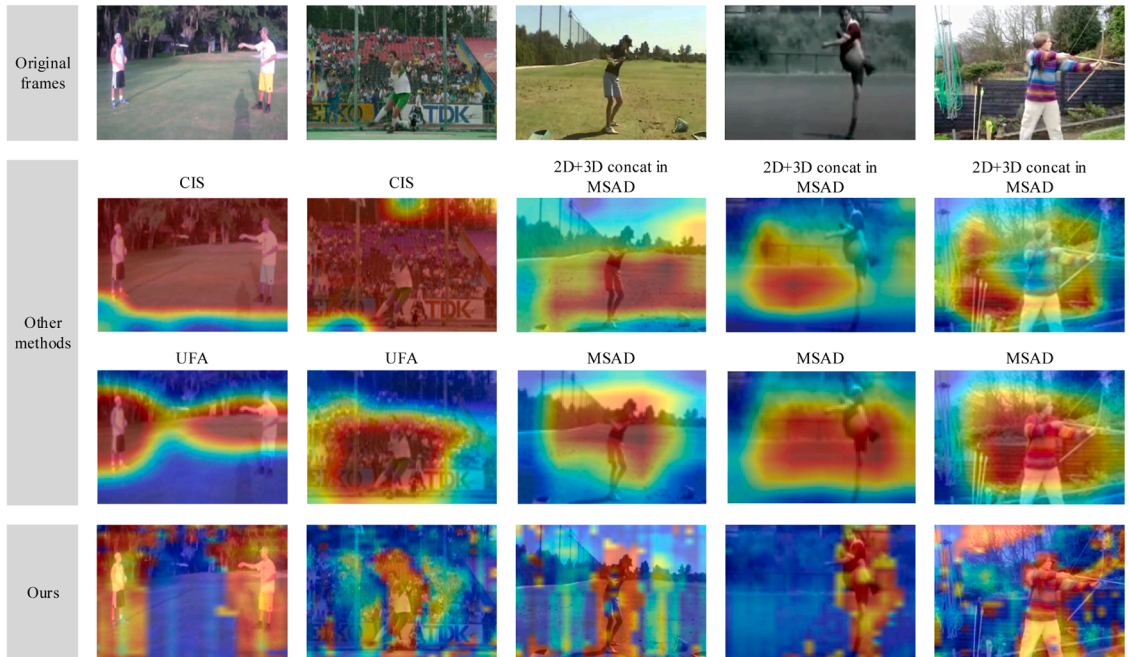


Fig. 8. Visualization results of the foreground attention map, where warm colors represent large weight values and cool colors represent low weight values. One frame was selected from each video to display the results. Among them, CIS denotes Context Information Separation, UFA denotes Unsupervised Foreground Attention and MSAD denotes Multi-scale Spatial Feature-aware Action Detection.

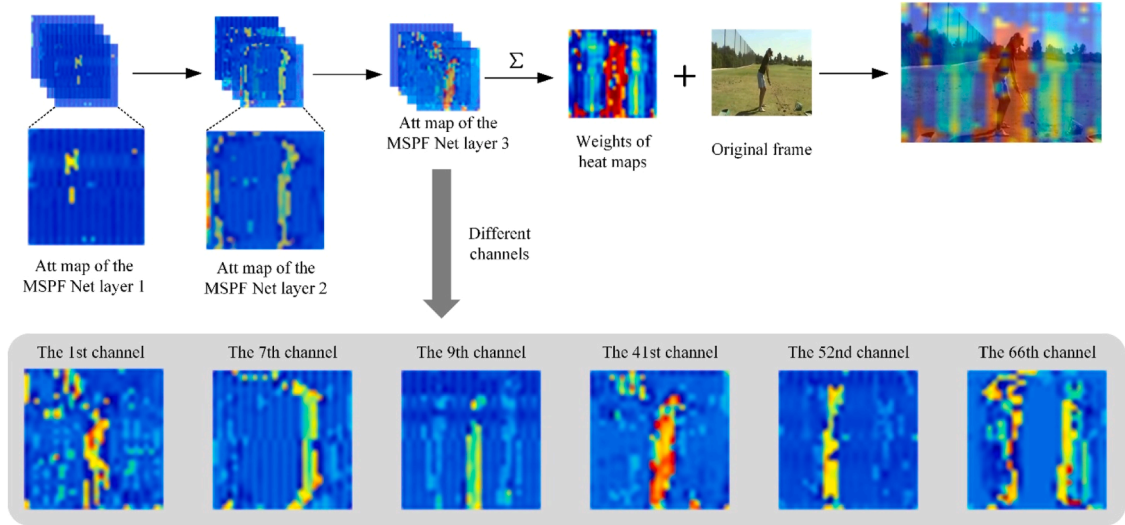


Fig. 9. Visualization results of attention maps for each layer in the Multi-Scale Pyramid-Former Network. Here, warm colors represent large weight values, while cool colors represent small weight values. The visualization results of the 1st, 9th, and 41st channels demonstrate the model's ability to focus on actions regions; the visualization results of the 7th, 52nd, and 66th channels show the model's ability to focus on action boundaries.

This not only enhances the model's perception of action boundaries but also facilitates the refinement of the model's detection results. The visualization results further validate the effectiveness of our method.

5. Discussion of results and implications

The proposed Multi-Scale Pyramid-Former Network with Multiple Consistency Constraints demonstrates significant advantages in semi-supervised action detection tasks, particularly excelling in scenarios with limited labeled data. Experimental results on the JHMDB-21 and UCF101-24 datasets indicate that MSPF Net requires only 20 % to 30 % of the annotated labels to achieve competitive accuracy. Specifically, on the JHMDB-21 dataset, MSPF Net surpasses state-of-the-art methods by 3.5 %, 0.7 %, and 3.1 % in terms of f-mAP@0.5, v-mAP@0.2, and v-mAP@0.5, respectively. On the UCF101-24 dataset, MSPF Net exceeds state-of-the-art methods by 0.7 %, 0.6 %, and 0.9 % for the f-mAP@0.5, v-mAP@0.2, and v-mAP@0.5 metrics, respectively.

Unlike existing SSVAD methods, which primarily focus on consistency constraints across different augmented data, these approaches have neglected modeling consistency across different scales of unlabeled video data. This oversight can lead to insufficient representation of unlabeled data by the model, resulting in overfitting and poor generalization. For example, ETEVAD (Kumar & Rawat, 2022) establishes consistency constraints on the localization results of videos before and after flipping; BWCC (Zhong et al., 2023) imposes consistency constraints on the prediction results of videos before and after background weakening; and Multi-FB (Zhang et al., 2025) enforces consistency constraints between foreground and background features in videos. In contrast to these approaches, our study argues that under semi-supervised conditions, relying solely on consistency constraints is insufficient to guide model training, as such constraints only enforce prediction stability on unlabeled data without ensuring the prediction precision. Therefore, under semi-supervised conditions, extracting richer scale features from unlabeled data is a promising approach. The proposed MSPF Net adaptively fuses information at different scales through the Pyramid Fusion Strategy, while establishing consistency constraints across various scales using the Coarse-to-Fine Multi-Scale Contrastive Strategy, thereby achieving more accurate detection results. Research on MSPF Net offers valuable insights for semi-supervised learning.

Furthermore, the proposed MSPF Net exhibits significant application potential in the medical and security fields. In medical scenarios, its multi-scale feature fusion capability can accurately capture organ details, and when combined with semi-supervised learning, it also effectively utilizes a large number of unlabeled clinical images, thereby alleviating the high cost of physician annotation. In security scenarios, MSPF Net reduces environmental interference by fusing multi-scale information and efficiently utilizes unlabeled surveillance video data based on the semi-supervised mechanism. These characteristics enable MSPF Net to closely align with practical application requirements. The method in this paper achieves advanced detection accuracy, thus being more suitable for scenarios requiring high model detection accuracy. However, it still has limitations: its relatively complex structure makes its deployment and debugging in practical applications relatively difficult.

6. Conclusion and future work

This study introduces a semi-supervised Multi-Scale Pyramid-Former Network with multiple consistency constraints. The implementation of this model consists of three steps: first, we output action features and category labels that focus on the global context through the Pose Encoder and Pose Activator. Then, we use the MSPF Net and Pyramid Fusion Strategy to fuse high-resolution global

features, local detail features, and low-resolution abstract semantic features at different scales, enabling the model to learn information at different scales and improve its representation capacity for unlabeled data. Finally, the Coarse-to-Fine Multi-Scale Contrastive Strategy is used to strengthen the consistency constraints of the Multi-Scale Pyramid-Former Network on unlabeled data. At the same time, the Pixel Domain Stepwise Localization Refinement Strategy enables the model to better discriminate between the action foregrounds and backgrounds in the video. We conducted extensive experiments on the benchmark datasets JHMDB-21 and UCF101-24, with results verifying that the proposed method achieved significant efficacy in semi-supervised scenarios while matching the performance of cutting-edge techniques in fully supervised settings.

However, our method performs poorly in scenarios with data imbalance. Specifically, in the semi-supervised training process, unlabeled data often have a class distribution bias. When the proportion of samples in some classes is too high, the model tends to be influenced by the features of large classes during classification, resulting in insufficient feature learning for small-sample classes. Thus, developing more targeted discriminative methods for imbalanced data scenarios is critical. We plan to continue our research on the aforementioned issues in future work.

CRedit authorship contribution statement

Qiming Zhang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Zhengping Hu:** Supervision, Funding acquisition, Formal analysis. **Yulu Wang:** Validation, Investigation, Data curation. **Hehao Zhang:** Project administration. **Jirui Di:** Resources.

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Data availability

Data will be made available on request.

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